

Department of Computing and Technology

IQRA University, H-9 Campus

Islamabad

Project Title: Title name as written in group list, shared with CR



Submitted by: Group Member 1 Name

Group Member 1 ID

Group Member 2 Name

Group Member 2 ID



Submitted to: Mr. Salman Ahmed

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1. Introduction

Provide a brief overview of the project, including its purpose and objectives.

Example:

Project Title: "Library Management System"

A Library Management System (LMS) is a software application designed to streamline the processes and operations of a library. It is used to manage and maintain the inventory of books, monitor transactions (such as borrowing and returning books), register users, and generate reports. By automating library functions, this system reduces manual errors, enhances efficiency, and provides users with a convenient way to interact with library resources.

Purpose of the Library Management System

The primary purpose of an LMS is to ensure the effective management of library resources, such as books, journals, magazines, and digital content. It aims to:

1. **Save Time:** Automating tedious tasks like cataloging books, tracking due dates, and maintaining user records saves significant time for library staff and users.
2. **Enhance Accuracy:** Eliminating manual processes reduces errors in book records, transactions, and user details.
3. **Improve Accessibility:** Users can easily search for books, check availability, and view their borrowing history without physically visiting the library.

Key Features of the System

1. **User Authentication:** Secure login for library administrators and members ensures data privacy.
2. **Book Management:** Allows adding, deleting, and updating book details, such as title, author, publication year, and availability status.
3. **User Management:** Registers new users, updates their details, and manages accounts of library members and staff.
4. **Borrowing and Returning Books:** Tracks the borrowing and return process, including due dates and overdue penalties.
5. **Search Functionality:** Users can search for books by title, author, or genre to check availability.

Benefits of the Library Management System

1. **Efficiency in Library Operations:** An LMS organizes and automates tasks such as tracking books, maintaining user records, and generating reports, allowing library staff to focus on more strategic functions.
2. **Enhanced User Experience:** Users can search, borrow, or renew books with ease, improving their interaction with the library.
3. **Cost-Effective Solution:** By reducing the reliance on paper records and manual efforts, an LMS minimizes operational costs.
4. **Data Security:** With user authentication and role-based access, the system ensures that sensitive data is protected.

2. Specifications

Technical Requirements:

- **Programming Language:** C++
- **Tools Used:** Code: Blocks or Visual Studio
- **Database:** File handling in C++
- **Platform:** Windows/Linux

Functional Requirements:

1. **User Login:** Authenticate users (admin or member) based on credentials.
2. **Dashboard:** Provide navigation to different modules.
3. **Book Management:**
 - Add, delete, and view book records.
4. **User Management:**
 - Register, delete, and view user details.
5. **Transactions:**
 - Borrow and return books.
 - Track due dates and generate fine reports.

3. Design / Architecture

Workflow Description:

1. The system begins with a **login screen**, verifying user credentials.
2. After successful login, the **dashboard** allows navigation between:
 - Managing books
 - Managing users
 - Borrowing/returning books
 - Generating reports
3. All data is stored in text files for persistence.

Flowchart:

- **Start**
- **Login Screen**
 - Valid User → **Dashboard**
 - Invalid User → **Exit**
- **Dashboard Options:**
 - **Manage Books** → Add/Delete/View Books
 - **Manage Users** → Register/Delete/View Users
 - **Borrow/Return Books** → Update Transactions
 - **Generate Reports** → Show Borrow Records
- **Logout**
- **End**

(Diagram must)

4. Implementation

Below is the **C++ code** implementing the described functionality:

5. Results and Analysis

Here, include screenshots of the program during runtime.

Example Screenshots:

1. **Login Screen:** A console view of the login screen.
2. **Dashboard:** A menu interface showing options for managing books and transactions.
3. **Book Management:** Screenshots showing the addition of a new book and the list of existing books.

6. References